

## PROJECT PRESENTATION

# Indoor Food Delivery Robot

An autonomous line-following robot that navigates predefined indoor routes, avoids obstacles, and shows live status on an onboard LCD — reducing manual effort and improving delivery accuracy.



## TEAM — SHORT CIRCUIT

- **Hamdil Hasan Partho** ID: 23101040
- **Sumaya Akter Lima** ID: 23101041
- **Jamshed Alam Sifat** ID: 23101055

# Problem Identification

Manual indoor food delivery creates repeated strain on staff and inconsistent service across busy dining spaces.



High workload for kitchen and floor staff



Delays in manual food delivery during peak hours



Risk of wrong-table delivery and mix-ups



Heavy dependence on staff availability



Busy, obstructed aisles and dining areas



Time lost on repetitive back-and-forth trips



Difficulty tracking order deliveries in real time



No consistent way to signal arrival at the table

# Our Solution

Mapping each delivery problem to a targeted automation response.

| PROBLEM                                     | SOLUTION                                              |
|---------------------------------------------|-------------------------------------------------------|
| High staff workload & repetitive trips      | Autonomous food-tray transportation                   |
| Delays in manual indoor delivery            | Automated point-to-point delivery on marked routes    |
| Risk of wrong-table delivery                | Predefined, 5-sensor line-guided routes to each table |
| Busy and obstructed dining aisles           | Real-time ultrasonic obstacle detection & stop        |
| No live delivery status for staff/customers | Onboard LCD shows movement, obstacle & arrival status |

# Project Objectives

Reduce manual delivery effort while making food service faster, safer, and more consistent through indoor automation.

01

## Reduce manual staff effort

Automate repetitive delivery trips across dining and workspace areas.

02

## Deliver orders more quickly

Cut delay between order-ready and table arrival with a direct route.

03

## Improve table-wise accuracy

Follow marked routes with 5-point IR sensing for precise drop-off.

04

## Enable safe, visible automation

Operate reliably around people, showing live status on the LCD.

# Key Features

Core capabilities that make indoor delivery autonomous, reliable, and visible.

01

Autonomous Navigation

02

5-Sensor Line Following

03

Ultrasonic Obstacle Detection

04

Junction-Based Table Stop

05

Live I2C LCD Status Display

06

Serial Diagnostics Monitor

# Operating Environment

Designed for reliable indoor delivery across shared indoor spaces.

**R**

## Restaurant

Follows marked routes through dining areas to reach assigned tables.

**U**

## University Campus

Handles changing foot traffic between rooms and common areas.

**O**

## Office Workspace

Moves quietly around staff and visitors during pantry-to-desk runs.

**C**

## Cafeteria

Navigates aisles and busy shared spaces to serve table-wise.

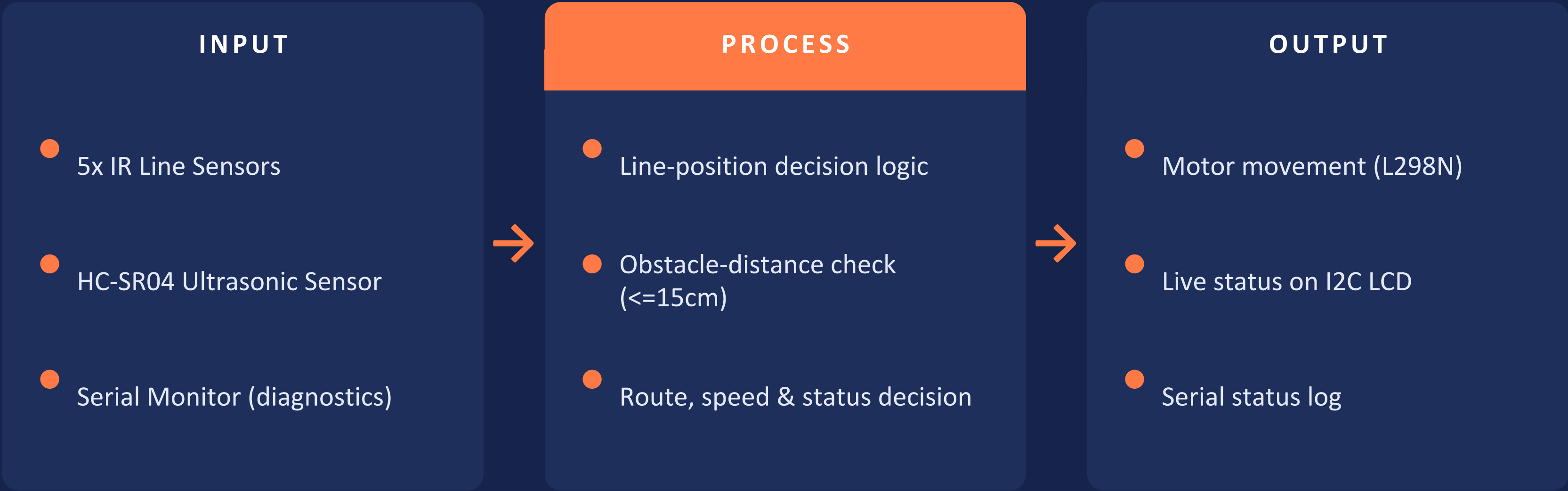
# Hardware Components

The robot combines sensors, control electronics, a status display, drive components, and a stable delivery platform.

| COMPONENT                   | ROLE                                                                       |
|-----------------------------|----------------------------------------------------------------------------|
| ESP32-S3 DevKit             | Main controller — runs sensor logic, motor control, and drives the I2C LCD |
| 5x IR Line Sensor Array     | Detects and tracks the guide line; all-HIGH pattern signals table arrival  |
| HC-SR04 Ultrasonic Sensor   | Measures distance ahead (up to ~400 cm) for obstacle avoidance             |
| 16x2 I2C LCD Display (0x27) | Shows live status: moving, turning, obstacle, or table reached             |
| L298N Motor Driver          | Controls direction and speed of both drive motors                          |
| 2x DC Geared Motors         | Drive the left and right chassis wheels                                    |
| Chassis & Delivery Tray     | Stable platform for carrying food orders                                   |
| 9V Battery Pack             | Portable power supply for all electronics                                  |

# Input, Processing & Output

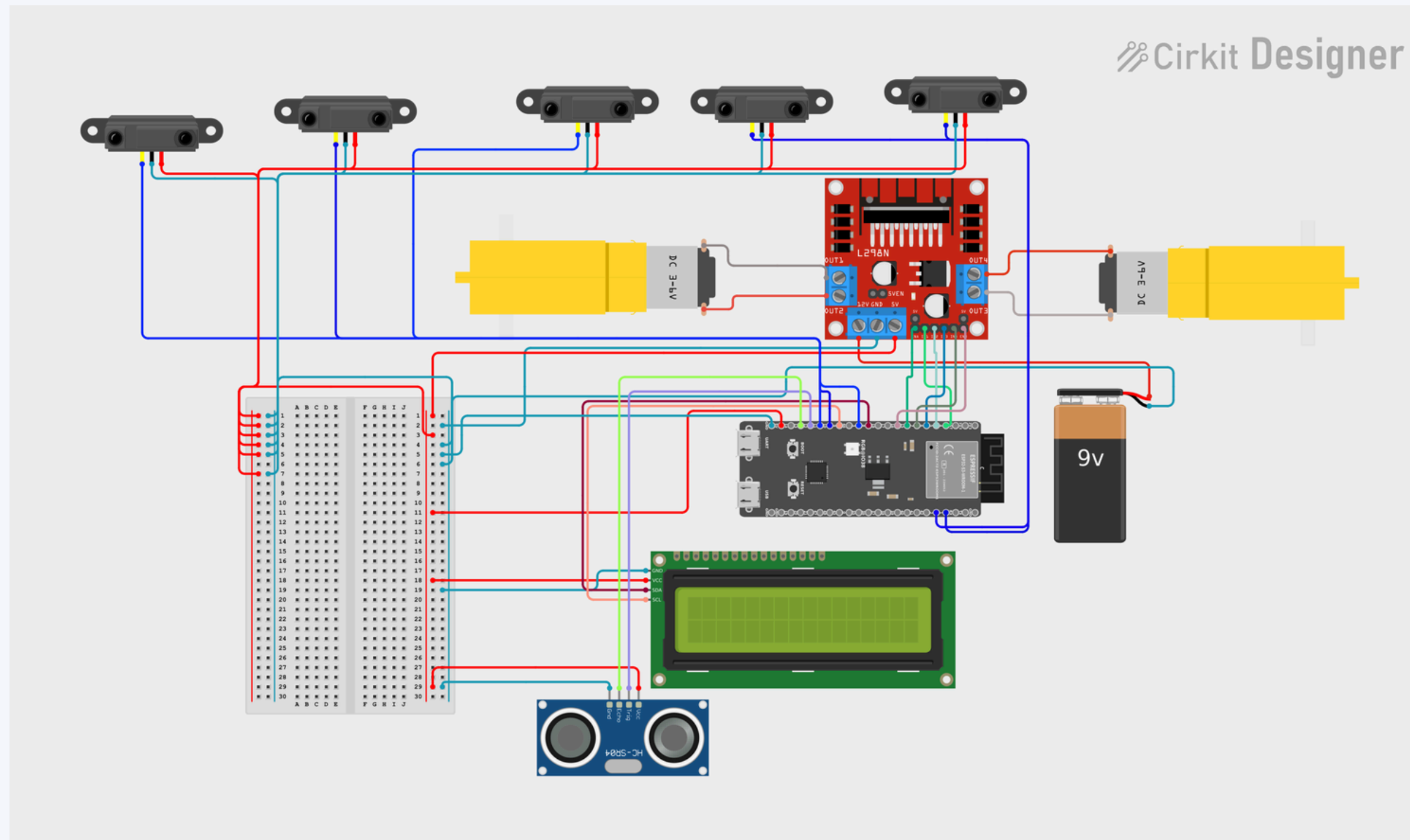
How sensor data becomes movement and live status updates.



*Decision-making layer converts sensor data into robot actions and human-readable status.*

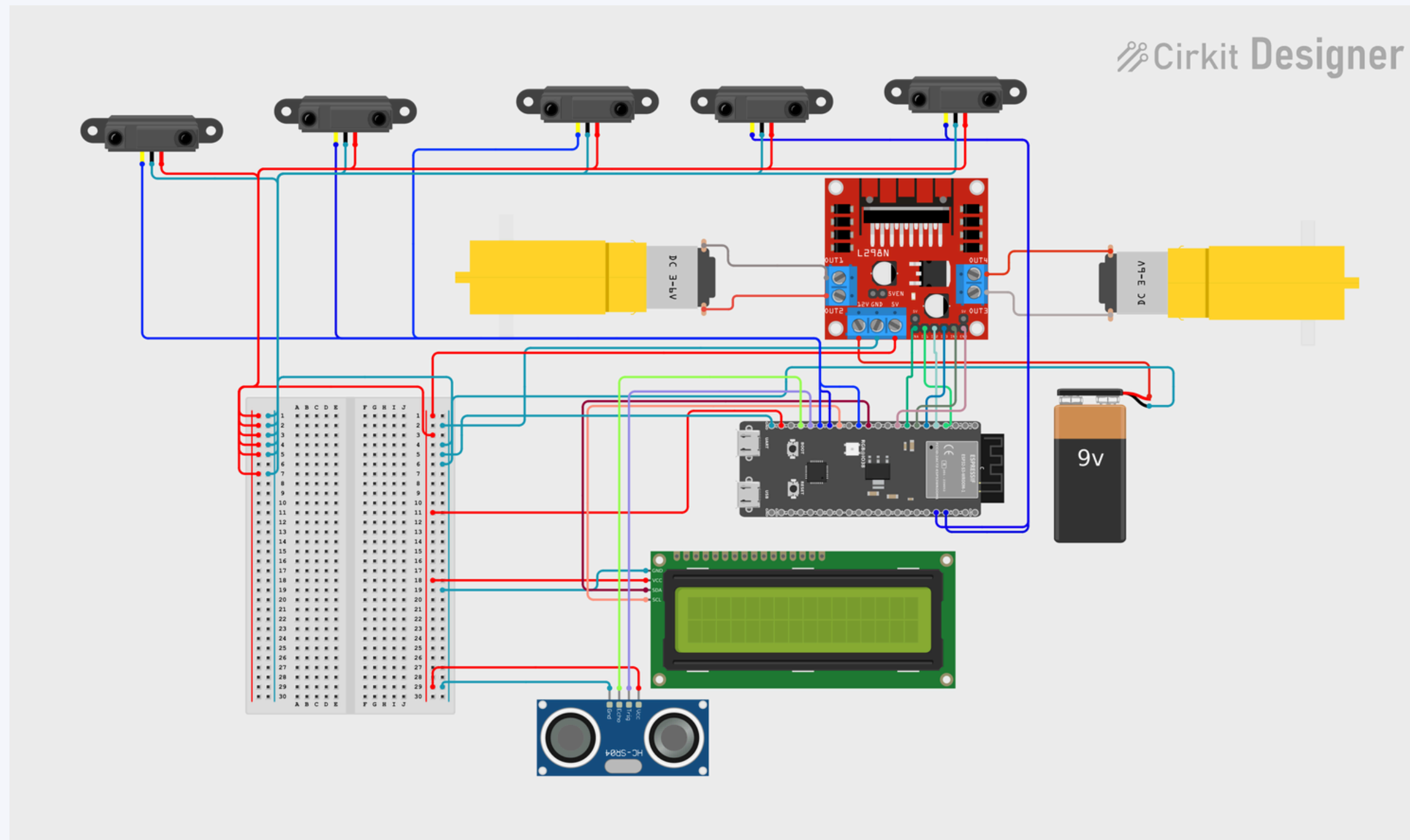
# Circuit Diagram

Updated wiring: ESP32-S3, 5x IR sensors, HC-SR04, L298N driver, dual DC motors, and I2C 16x2 LCD.



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# Line-Following & Table Detection

The 5-sensor IR array keeps the robot centered on the guide line and detects table arrival at a junction.

## ● ● ● PSEUDOCODE — LINE FOLLOWING (5x IR SENSOR ARRAY)

```
START
Initialize 5 IR sensor pins (FL, L, C, R, FR) as INPUT
Initialize L298N motor pins as OUTPUT

WHILE robot is ON:
  Read FL, L, C, R, FR sensor values
  IF all 5 sensors are HIGH THEN
    STOP motors    // junction / table reached
  ELSE IF only C is HIGH THEN
    Move forward
  ELSE IF L and C are HIGH THEN    Turn left (correct)
  ELSE IF R and C are HIGH THEN    Turn right (correct)
  ELSE IF FL or L is HIGH THEN     Sharp turn left
  ELSE IF FR or R is HIGH THEN     Sharp turn right
  ELSE
    STOP motors    // no line detected, search
  END IF
END WHILE
END
```

# Obstacle Detection & Avoidance

The HC-SR04 ultrasonic sensor continuously checks the path ahead and halts the robot before a collision.

## ● ● ● PSEUDOCODE — readUltrasonicDistance()

```
FUNCTION readUltrasonicDistance():
  Set TRIG pin LOW, wait 2 microseconds
  Set TRIG pin HIGH, wait 10 microseconds, then LOW
  Measure ECHO pulse duration (30 ms timeout)
  IF no echo received THEN
    RETURN 999    // out of range, > 400 cm
  ELSE
    RETURN distance = duration * 0.034 / 2    // cm
  END IF

MAIN LOOP:
  distance = readUltrasonicDistance()
  IF 0 < distance <= SAFE_DISTANCE (15 cm) THEN
    STOP motors
    Show "OBSTACLE" + distance on LCD
  ELSE
    Continue line-following logic
  END IF
```

# Motor Control (L298N Driver)

Drive logic converts a movement decision into direction and speed signals for both DC motors.

## PSEUDOCODE — MOTOR CONTROL FUNCTIONS

```
CONST MOTOR_SPEED = 180    // of 255, ~70% duty cycle

FUNCTION moveForward():
  Set ENA, ENB = MOTOR_SPEED
  IN1=HIGH, IN2=LOW        // left motor forward
  IN3=HIGH, IN4=LOW        // right motor forward

FUNCTION turnLeft():
  IN1=LOW, IN2=HIGH        // left motor reverse
  IN3=HIGH, IN4=LOW        // right motor forward

FUNCTION turnRight():
  IN1=HIGH, IN2=LOW        // left motor forward
  IN3=LOW, IN4=HIGH        // right motor reverse

FUNCTION stopMotors():
  ENA=0, ENB=0
  IN1=IN2=IN3=IN4=LOW
```

# Live Status on I2C LCD

A 16x2 I2C LCD replaces indicator LEDs, showing the robot's current action in real time.

## ● ● ● PSEUDOCODE — LCD STATUS DISPLAY (I2C, addr 0x27)

```
START
Initialize I2C (SDA = GPIO 8, SCL = GPIO 9)
Initialize LCD at address 0x27, backlight ON

WHILE robot is ON:
  IF obstacle detected THEN
    Line 1: "Status: OBSTACLE"
    Line 2: "Dist: <value> cm"
  ELSE IF at junction THEN
    Line 1: "Status: JUNCTION"
    Line 2: "Table Reached"
  ELSE IF following line THEN
    Line 1: "Status: MOVING"
    Line 2: "Path: FORWARD / LEFT / RIGHT"
  ELSE
    Line 1: "Status: SEARCHING"
    Line 2: "No Line Found"
  END IF
END WHILE
```

# Controller & Responsibilities

The ESP32-S3 runs all sensing, decision, motor, and LCD logic while offering built-in wireless connectivity.

## ESP32-S3

DevKit — dual USB (UART + native USB), 5x IR, HC-SR04, L298N, and I2C LCD all wired directly to its GPIOs.

**I2C bus** SDA = GPIO 8, SCL = GPIO 9

**Serial monitor** 115200 baud diagnostics

**Loop timing** 50 ms per control cycle

### CONNECTIVITY

Built-in Wi-Fi and Bluetooth for future remote monitoring and OTA updates.

### ROLE

Single controller handles sensing, navigation logic, motor driving, and the LCD.

### KEY BENEFIT

Wireless-ready hardware keeps navigation precise today and smart-integration ready tomorrow.

# Measurable Parameters

Quantifiable thresholds and targets used to tune and evaluate the robot.

|                                                                                                                        |                                                                                                                            |                                                                                                                  |                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <div>15 cm</div> <div>Obstacle safe distance</div> <div>Robot stops when an object is within this range</div>          | <div>180 / 255</div> <div>Motor drive speed (PWM)</div> <div>≈ 70% duty cycle for smooth, controlled motion</div>          | <div>≈ 400 cm</div> <div>Ultrasonic max range</div> <div>Beyond this, sensor reports out-of-range (999)</div>    | <div>20 Hz (50 ms)</div> <div>Control loop refresh rate</div> <div>Sensor read → decision → action cycle time</div> |
| <div>5 channels</div> <div>Line sensor array</div> <div>FL, L, C, R, FR — enables sharp &amp; gentle corrections</div> | <div>5/5 sensors HIGH</div> <div>Table-arrival detection</div> <div>All sensors on the line marks a junction / table</div> | <div>115200 baud</div> <div>Serial diagnostics rate</div> <div>Real-time sensor &amp; motor status logging</div> | <div>≥ 95%</div> <div>Target delivery accuracy</div> <div>Correct-table delivery rate across test runs</div>        |

# Budget Estimation

Approximate hardware cost based on shared component pricing. Figures are estimates and may vary by supplier.

| COMPONENT                  | QTY | EST. COST (BDT) |
|----------------------------|-----|-----------------|
| ESP32                      | 1   | 900             |
| IR Sensor Array (5-way)    | 1   | 200             |
| HC-SR04 Ultrasonic Sensor  | 1   | 120             |
| 16x2 I2C LCD Display       | 1   | 280             |
| L298N Motor Driver         | 1   | 180             |
| DC Geared Motors * 2       | 2   | 295             |
| Chassis & Delivery Tray    | 1   | 1100            |
| 9V Battery + Wires & Misc. | —   | 300             |
| ESTIMATED SUBTOTAL         |     | 5375 BDT        |

## WHAT CHANGED

Controller upgraded to ESP32-S3 and a 16x2 I2C LCD was added for live status — replacing the earlier LED-indicator plan.

# Expected Benefits

Automated delivery helps staff save time, improves consistency, and creates a safer, more visible workflow.

01

## Accurate table-wise delivery

Meals reach the right table using 5-sensor marked routes.

02

## Reduced manual effort

Staff spend less time on repetitive delivery runs.

03

## Safe indoor operation

Ultrasonic sensing keeps movement collision-free.

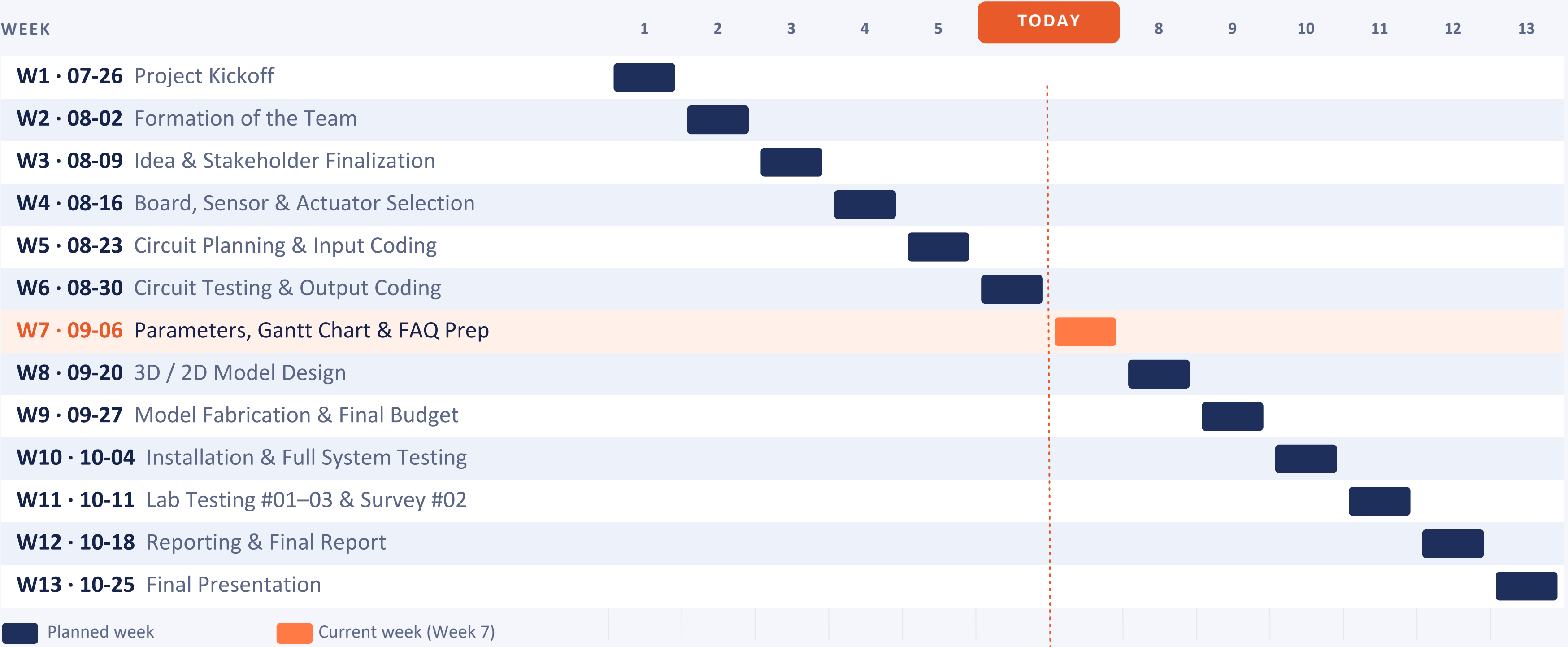
04

## Real-time visibility

LCD status keeps staff informed without guesswork.

# Gantt Chart

13-week execution plan, 26 Jul – 25 Oct 2026 — currently in Week 7.



# Team Operations Management

| Member                         | Role                                           | Key Responsibility                                                                                                                                                             |
|--------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hamdil Hasan Partho (23101040) | Team Lead / Hardware & Circuit Assembly        | Project coordination; circuit design & assembly; 5-sensor IR array & HC-SR04 integration; L298N motor driver & motor setup; hardware testing & calibration; budget estimation. |
| Sumaya Akter Lima (23101041)   | Documentation & Presentation / Project Support | Stakeholder survey design & reporting; report writing; presentation slide design; team operations management; budget tracking; QA support.                                     |
| Jamshed Alam Sifat (23101055)  | Firmware / Software Development                | ESP32-S3 programming; line-following & obstacle-detection logic; I2C LCD status-display coding; code testing & debugging; system integration testing.                          |

# Initial FAQ

Anticipated questions for the project defense.

**Q Why did you choose line-following over full SLAM/mapping navigation?**

**A** Line-following is low-cost, reliable indoors, and easy to redeploy — no expensive LIDAR or mapping compute needed.

**Q How does the robot know it has reached the correct table?**

**A** All 5 IR sensors reading HIGH together marks a junction on the guide tape, which is placed at each table stop.

**Q What happens if a person steps in front of the robot?**

**A** The HC-SR04 sensor stops the motors the moment distance falls to 15 cm or less, and the LCD shows an obstacle alert.

**Q Why switch from LED indicators to an LCD screen?**

**A** A 16x2 LCD shows richer, human-readable status (moving, turning, obstacle, arrived) instead of just a single color code.

**Q What is the control loop's response time?**

**A** The main loop runs roughly every 50 ms (~20 Hz), giving fast enough reaction for typical indoor walking speeds.

# Initial FAQ

Anticipated questions for the project defense.

**Q Why ESP32-S3 instead of a basic Arduino board?**

**A** It offers more GPIO, built-in Wi-Fi/Bluetooth for future remote monitoring, and enough performance for all sensors plus the LCD.

**Q How is speed controlled on the motors?**

**A** The L298N driver receives a PWM value (180 of 255) on ENA/ENB, giving smooth, consistent forward motion.

**Q What happens if the robot loses the line entirely?**

**A** It stops immediately and the LCD shows 'No Line Found' rather than moving blindly in an unknown direction.

**Q Can the robot serve more than one table per trip?**

**A** The current build handles one route to one destination; multi-stop routing is listed under future scope.

**Q How much does the hardware cost in total?**

**A** Roughly 3,540 BDT based on current component pricing, detailed on the Budget Estimation slide.

# Survey#01 Reporting



# Design Simulation Link

Cirkit Designer

# Practical Automation & Future Scope

A practical foundation for smarter indoor food delivery.

## PROJECT OUTCOME

- Practical indoor automation
- Table-wise delivery accuracy
- Live LCD status visibility
- Reduced manual staff effort
- Useful across indoor spaces

## FUTURE DIRECTIONS

### Smart connectivity

Remote monitoring via the ESP32-S3's Wi-Fi/Bluetooth

### Route expansion

Support multiple tables and destinations

### Adaptive navigation

Improve routing for changing layouts

### Data insights

Track deliveries for better service planning



# Thank You

QUESTIONS ARE WELCOME

Thank you for your attention. We welcome your questions and discussion about our indoor food delivery robot.